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Sexual dimorphism in the microbiology of the CF 'Gender Gap': Estrogen modulation of *Pseudomonas aeruginosa* virulence

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ABSTRACT

There is increasing evidence for sexual dimorphism of estrogen (E2) actions in the exacerbation of lung function, infection and inflammation in females with cystic fibrosis - the so-called "CF gender gap". The effects of estrogen on virulence factors that enhance *P. aeruginosa* persistence in CF lung epithelium were investigated by phenotypic and chemical assays in various *PsA* clinical isolates and laboratory strains in isolation or in co-culture with normal (Nuli-1) and CF dPhe508-CFTR (CuFi-1) human bronchial epithelial cell lines. Estrogen (E2, 10 nM) significantly increased secretion of the virulence factor pyocyanin by 80% in *PsA* early infection isolates from female CF patients and by 280% in late infection *PsA* isolates. Estrogen also increased the swarming motility by up to 50% in all *PsA* isolates and strains tested in 0.5% agar. A significant increase of 110% in the twitching motility of all *PsA* isolates and strains tested was also observed with estrogen treatment. Treatment with E2 increased biofilm formation of *P. aeruginosa* PsAO1 which became more adherent to, and invasive into, normal and CF bronchial epithelial cells. The selective estrogen receptor modulators (SERMs), Tamoxifen and ICI 182780 inhibited *P. aeruginosa* motility. The potency of various steroid hormones to stimulate motility of P. aeruginosa was in the order; estradiol ≫ estrone > E3 estriol ≥ testosterone ≥ progesterone ≫ aldosterone, cortisol. Estrogen was also shown to reduce ciliary beat intensity in CF bronchial epithelium which would further exacerbate PsA trapping and virulence in the CF airways. In conclusion, we have demonstrated for the first time that estrogen exacerbates *P. aeruginosa* virulence factors and enhances bacterial interactions with CF bronchial epithelium which can be inhibited by tamoxifen. Our work suggests that SERMs could be used as an adjuvant treatment to reduce estrogen-induced *P. aeruginosa* infections and associated lung exacerbations in females with CF.

1. Introduction

Cystic fibrosis (CF) is an autosomal genetic disorder and is the most common lethal genetic disease among Caucasians, affecting approximately 1:2500 people worldwide [1]. There is a reported survival disadvantage for women [2] and significant sexual dimorphism has been reported in the progression of CF in male and female patients – commonly (and erroneously) cited in the literature as "the CF gender gap" [3]. Male CF patients survive on average five to ten years longer than females [4] and lung exacerbations in CF females have earlier onset at puberty and vary during the estrous cycle [5].

We have previously demonstrated a role for estrogen in the exacerbation of lung function, infection and inflammation, and airway surface liquid dynamics in females with cystic fibrosis [6,7]. Estrogen reduces airway surface liquid height in normal airway epithelium principally by inhibiting wild-type CFTR and further reduces ASL in CF epithelium by activation of the epithelial Na^+ channel ENaC [7]. Although these different strands of evidence point to a pro-virulence effect of estrogen in the CF airway, the underlying molecular mechanisms are poorly understood, particularly for the microbial endocrine effects of estrogen on *Pseudomonas aeruginosa* (*PsA*). Here we report for the first time the direct effects of estrogen on *PsA* virulence factors and invasion.

17β-estradiol (E2) is the most potent circulating estrogen hormone during the reproductive years in women. Female gender enhances the risk of worse prognosis in pulmonary diseases such as CF and the role of E2 in lung physiology and pathophysiology is still a matter of debate. From epidemiological studies it appears that estrogen is deleterious to

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female CF lung function and the deterioration of the CF condition seen in girls at puberty has been attributed to an elevation of estrogen levels in the body [3]. We previously reported that exacerbations in lung function in CF females were associated with higher levels of estrogen during the menstrual cycle [6] and demonstrated that estrogen activation of the Na^+ channel ENaC in CF bronchial epithelium decreased the height of the airway surface liquid (ASL) layer [7]. CF lung infections also display marked sexual dimorphism and female CF patients develop more serious infections with *Burkholderia* spp. and their lungs are colonized earlier with *P. aeruginosa* [8].

P. aeruginosa is known to employ numerous virulence factors during both acute and chronic colonization of the CF lung which promote infection, as reviewed by Winstanley et al. [9]. These acute virulence factors include rhamnolipids, endotoxins and quorum sensing systems which control important secreted virulence factors such as pyocyanin and elastase and phenotypes such as motility. Chronic virulence factors include siderophores, a switch from mucoid to non-mucoid phenotype, biofilm formation, cell adhesion and invasion.

In recent years, a number of studies have reported an effect of estrogen on *P. aeruginosa* both *in vitro* and *in vivo*. We have found that estrogen induces mucoid conversion of *P. aeruginosa* in females with CF through a mutation of *mucA* [6] and also supresses lactoferrin, which is known to prevent biofilm formation [10]. Although these different strands of evidence point to a pro-virulence effect of estrogen in the CF airway, the underlying molecular mechanisms are poorly understood, particularly for the microbial endocrine effects of estradiol on *P. aeruginosa*. This study investigates the effect of estrogen on *P. aeruginosa* virulence factors that enhance colonization of the CF lung in a bid to further understand the role and sexual dimorphism of estrogen in the exacerbation of CF airway disease.

2. Methods

2.1. Chemicals

17β-estradiol and 4-hydroxy tamoxifen (cell culture tested) were purchased from Sigma (Sigma-Aldrich, Dublin, Ireland). ICI 182780, diarylpropionitrile (DPN), and pyrazole triol propyl (PPT) were purchased from Tocris Bioscience (Bristol, UK). All above chemicals were dissolved in ethanol and used at a final concentration as indicated in the text. All other chemicals were purchased from Sigma-Aldrich.

2.2. Bacterial strains and clinical isolates

P. aeruginosa strain PAO1 was kindly provided by Professor Fergal O'Gara, University College Cork, Ireland. *P. aeruginosa* strain 27853 was purchased from American Type Culture Collection (ATCC). Clinical isolates of *P. aeruginosa* were isolated from BAL specimens of two preschool children with CF at Our Lady's Children's Hospital Crumlin, Dublin, Ireland (Local Ethics Committee approval). One isolate was from the early stage of infection from one patient and a second isolate was from a later stage of infection in a second patient. These isolates are not clonal variants from the same patient. The *P. aeruginosa* isolates had a non-mucoid phenotype and are broadly susceptible to anti-pseudomonal antibiotics. Two lab strains and two clinical isolates were used in this study to observe if there were differences between lab and clinical strains of *P. aeruginosa* after estrogen treatment. All bacterial strains were maintained on LB agar and either grown in LB broth or minimal growth medium (MM9) at 37°C for experiments.

2.3. Polarized cell culture

The host-pathogen studies were carried out on two airway epithelial cell lines: a normal bronchial epithelial cell line NuLi-1 derived from normal human airway epithelium of normal genotype, and a CF cell line, CuFi-1 derived from bronchial epithelium of a homozygous CFTR F508del/F508del individual. NuLi-1 and CuFi-1 were generous gifts from Drs A. Klingelhutz, P. Karp, and J. Zabner (University of Iowa, Iowa City, IA). Cells were initially grown to confluency in human placental collagen type VI (Sigma-Aldrich)-coated flasks using Bronchial Epithelium Basal Medium (BEBM, Clonetics™, Lonza, Isis, Bray, Ireland) supplemented with 0.1% human recombinant epidermal growth factor, 0.4% bovine pituitary extract, 0.1% epinephrine, 0.1% transferrin, 0.1% insulin, 0.1% retinoic acid, 0.1% hydrocortisone, 0.1% triiodothyronine, and 50 μg/ml gentamicin (Gibco®, Invitrogen, Dun Laoghaire, Ireland), 0.025 μg/ml penicillin/streptomycin and 1.25 μg/ml Fungizone® Gentamicin (Gibco®). These cell lines were then grown in a liquid/liquid interface on human placental collagen type VI Millicell hanging cell culture inserts (Millipore, Billerica, USA) in BEGM. Once confluence was reached, cells were cultured in a 1:1 ratio of Dulbecco's Modified Eagle's Medium (DMEM) and Ham's F-12 Medium (Sigma-Aldrich) supplemented with 2% UltroserG (Biosciences, Dun Laoghaire, Ireland), 50 units-50 μg/ml penicillin/streptomycin, 50 μg/ml gentamycin and 1.25 μg/ml amphotericin B. After 2 days in this DMEM:Ham's F12 media, the cells were cultured under an air/liquid interface. Transepithelial electrical resistance (TEER) measurements were taken every 4–5 days using the Epithelial Volt-Ohm Meter (EVOM, World Precision Instruments, Aston, UK) according to manufacturer's instructions and converted to $\Omega.cm^{-2}$ based on surface area of cell culture insert. Once TEER reached at least 1000 $\Omega.cm^{-2}$ (indicating the formation of a well differentiated epithelium - approximately 28 days), cells were used for experiments.

NuLi-1 and CuFi-1 were grown as epithelial monolayers on a collagen coated fibronectin support in miniature Ussing chambers under thin film culture conditions. The NuLi-1 and CuFi-1 airway epithelial monolayers were tested for normal and CF phenotype by measuring the airway surface liquid height using Dextran coupled Texas Red to stain the hydrated ASL layer and calcein green to stain cells [7]. The NuLi-1 and CuFi-1 epithelia exhibited ASL dynamics characteristic of normal and CF airway (Supplemental Fig. S1) which reflect the ion transport characteristics of these epithelia.

2.4. Phenotypic assays

2.4.1. P. aeruginosa growth curve analysis

To evaluate the effect of E_2 on *P. aeruginosa* growth, a growth curve analysis was conducted over an 8 h period. Growth curve analysis did not reveal a significant difference in the growth of any of the *P. aeruginosa* isolates treated with E_2 compared to vehicle control treated bacterial cultures (Supplemental Fig. S2). Similar results were found for longer periods of E2 incubation over a 24 h period. Routinely, 24 h cultures of *P. aeruginosa* were used to prepare an inoculum of 1:100 in conical flasks containing 30 ml LB media, ± 10 nM E_2 or ethanol (vehicle control). Cultures were grown at 37°C and samples were taken at 2, 4, 6, 8 and 24 h. OD was measured at $OD_{600\ nm}$ and CFU/ml was quantified by viable counts on LB agar.

2.4.2. Pyocyanin quantification assay

Pyocyanin concentrations were measured by a quantitative chemical assay. Overnight cultures of *P. aeruginosa* grown in LB media were adjusted to an $OD_{600\ nm}$ of 0.2 in LB media, ± 10 nM E2 or ethanol (vehicle control). After 24 h incubation, 2 ml of cell-free supernatant from the cultures was extracted with 1 ml of chloroform and re-extracted into 0·3 ml of 0·2 mol l^{-1} HCl to obtain a pink to deep red solution [11]. The pyocyanin concentration in this solution was measured at $OD_{520\ nm}$. The concentration of pyocyanin in the supernatant was expressed in μg/mL by multiplying the absorbance by 17.072.

2.4.3. Total protease activity assay

Total protease activity was determined with an azocasein assay as previously described [12]. Overnight cultures of *P. aeruginosa* grown in LB media were adjusted to an $OD_{600\ nm}$ of 0.2 in LB media, ± 10 nm E2

or ethanol (vehicle control). After 6 h of growth, the supernatant was collected and filter-purified using a 0·22-μm nylon filter, and 100 μl of the filtered culture supernatant was added to a reaction buffer (400 μl of 50 mmol l^{-1} K_2HPO_4 (pH 7·0) containing 0·8% azocasein. The reaction mixture was incubated at 30C for 3 h. After the addition of 400 μl of 1·5 mol l^{-1} HCl and incubation on ice for 30 min, the mixture was centrifuged at 7620*g* for 10 min, and 0·5 ml of 1 mol l^{-1} NaOH was added to the supernatant. The absorbance of the clear supernatant was measured spectrophotometrically at $OD_{440\ nm}$.

2.4.4. Siderophore production

The liquid Chrome Azurol S (CAS) assay was carried out as described in [13]. Overnight cultures of *P. aeruginosa* grown in MM9 media were adjusted to an $OD_{600\ nm}$ of 0.2 in MM9 media, ± 10 nM E_2 or ethanol (vehicle control). Supernatants were filtered using 0.22 μM filters to remove cells. A 500 μl volume of cell-free supernatant was mixed with 500 μl CAS solution (150 μM CAS, 15 μM $FeCl_3$, 0.5 M Piperazine, 1.2 mM HDTMA) and 10 μl shuttle solution (0.2 M 5-Sulfosalicylic acid) and incubated for 5 h at room temperature. Reference samples of bacterial growth medium containing the relevant concentration of iron were also assayed. Absorbance was measured at $OD_{630\ nm}$ using MM9 alone as a blank. The percentage siderophore units were calculated as $[(A_r - A_s)/A_r] \times 100$ where A_r is the absorbance of the reference and A_s is the absorbance of the sample.

2.4.5. Motility assay

Bacteria from overnight *P. aeruginosa* cultures in LB broth were inoculated on the surface of 0.25% (swimming), 0.5% (swarming) and 1.0% (twitching) agar ± 10 nM E2, Tamoxifen (30 μM), ICI 182, 780 (0.5 nM), DPN (5 nM) or PPT (15 nM) or appropriate vehicle controls and incubated at 37C for 24 hr. The motility ability was assessed by examining the movement distance of the bacteria away from the inoculation point. An estrogen binding protein (EBP receptor) has been previously characterized in the cytosol of PsA with high binding affinity for E2 compared to other steroids [4]. We compared the steroid binding selectivity of the EBP receptor, calculated from *PsA* EBP binding data in Rowland SS et al. [24], with the potency to stimulate PsA motility of E2 and other estrogens (estrone), (estriol) and other steroid hormones, testosterone, progesterone, aldosterone and cortisol, all added at the same concentration (10 nM) for similar incubation times.

2.4.6. Crystal violet biofilm assay

Biofilm formation was assessed in 96-well PEG lid MBEC assay plate (Innovotech). Briefly, overnight cultures of *P. aeruginosa* grown in LB media were adjusted to an $OD_{600\ nm}$ of 0.2 in LB media, ± 10 nM E2 and testosterone or ethanol (vehicle control). Plates were incubated for 72 h at 37°C with light shaking. Thereafter, non-adherent bacteria were removed by washing with sterile phosphate-buffered saline (PBS), and adherent bacteria were stained for 15 min with a 1% crystal violet solution. Pegs were washed with PBS to remove excess stain. After the wells had air-dried, the crystal violet bound to the biofilm was extracted with 95% ethanol. The absorbance of the crystal violet solution was measured spectrophotometrically at $OD_{590\ nm}$. Wells containing medium were used as blank controls.

2.4.7. Gentamicin exclusion assay

P. aeruginosa was grown in LB overnight shaken at 37°C. The bacterial solution was diluted 1:20 in BEGM bronchial epithelium medium without antibiotics and grown to log phase at $OD_{600\ nm}$. Upon reaching log phase, bacteria were added to fresh BEGM/DMEM bronchial media ± 10 nM E2 or ethanol (vehicle control) or bacteria and cells were treated separately ± E2 or ethanol (vehicle control) for 72 h prior to infection. Then, 25 μl of the bacterial suspension was added apically for 2h on NuLi-1 and CuFi-1 cells at a MOI of ~1 in 10. Following bacterial incubation, the epithelial cells were washed and treated apically with gentamicin for 1 h to kill any extracellular bacteria for quantification of internalized bacteria. For quantification of adherent bacteria treatment cells were not treated with gentamicin. Epithelial cultures were washed twice with PBS and treated with Triton-X 100 (0.4% v/v) for 20 min to lyse epithelial cells and release internalized bacteria. Lysates were serially diluted and plated on LB agar plates and incubated at 37°C overnight. CFU/ml was recorded (accurate range ≥ 30–≤300).

2.4.8. Ciliary beat intensity

Estrogen may affect *PsA* virulence in CF airways by reducing mucociliary clearance. Estrogen reduces ASL height in CF airway epithelium below the level of outstretched cilia [7] which in itself would inhibit ciliary beating and mucociliary clearance. We determined if E2 could act directly on ciliary beat independently of changes in ASL height by measuring ciliary beat intensity in CuFi-1 cells grown as monolayers as described above except in this case the cell culture well was flooded to raise the ASL height 1 mm above the cell layer for 1 h prior to treatment with E2 (1 nM). Single cell ciliary beating was measured in NuLi-1 and CuFi-1 epithelial monolayers at 37°C using differential interference contrast (DIC) on a Zeiss CLSM 510 inverted microscope. Cilia beating images were acquired for regions of interest size 64 × 64 pixels using an Andor iXon® Ultra 897 high-speed digital imaging camera (at 280 frames. s^{-1}) operating MetaMorph® 7.7 software. Images were analysed using ImageJ software® and ciliary beat intensity calculated as the product of ciliary beat frequency (CBF) and peak to peak amplitude (CBA) of waves of ciliary beating measured from the changes in DIC optical density (OD) created by the ciliary beat. Peak-to-peak amplitude was measured from the change between peak OD (highest amplitude value) and trough OD (lowest amplitude value) of the ciliary beat oscillations. The cilia appear as a tuft of approx. 6–8 μm in height on top of single cells. Ciliary beat was detected from variations in the grey intensity of the DIC image that resulted from the repetitive oscillations of the cilia and analyzed by selecting regions of interest (68 × 68 pixels) near the midline of the cilia. The ciliary beat frequency was calculated from the period (frequency in hertz = 1/period) of each cycle of the grey-intensity oscillations by beat-by-beat analysis. A high-speed camera is required to accurately detect the CBF rates of 20 Hz. CBF was relatively stable for recording periods of 30 min as long as the epithelium was maintained at 37°C. Reducing the incubation temperature to 20°C greatly attenuated the CBF by up to 50% and caused a progressive decline of CBF during recordings of duration of more than 5 min.

2.4.9. Statistical analysis

Data shown are means ± SEM, and the Student paired *t*-test were used to compare the means of different treatment groups unless otherwise indicated. Unpaired *t*-test of treatments vs. vehicle controls and NuLi-1 vs. CuFi-1 cells. Statistical significance was assessed by ANOVA. The difference between two means was considered to be significant when $p < 0.05$. All experiments were preformed in triplicate.

3. Results

3.1. Estrogen increases pyocyanin secretion by P. aeruginosa

To determine if estrogen has an effect on secreted *P. aeruginosa* virulence factors, *PsA* isolates were grown in the presence of E2 or vehicle control for 24 h and assessed for pyocyanin, protease and siderophore secretion. E2 significantly increased the secretion of pyocyanin, as determined by a quantitative chemical assay, in the two *P. aeruginosa* clinical CF isolates. There was a 79% increase in pyocyanin secretion in the early isolate ($P < 0.05$) and a 279% increase in secretion in the late isolate ($P < 0.01$) (Fig. 1A) compared to vehicle controls. There was no significant difference in pyocyanin secretion in *P. aeruginosa* 27853 or PAO1 with E2 treatment compared to vehicle controls (Fig. 1A). For all isolates tested, treatment with E2 did not

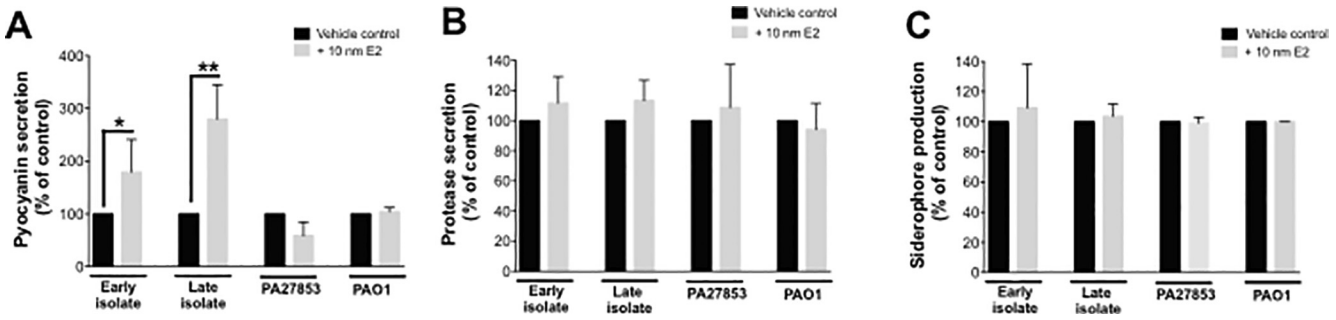

Fig. 1. Effects of 10 nM E2 on *P. aeruginosa* virulence factors. (A) pyocyanin, (B) protease, (C) siderophore secretion. The results are presented as the mean ± SE of the percent absorption at different wavelengths with respect to the corresponding control. Significance *P < 0.05 (vs vehicle control).

reveal any significant difference in total protease activity or siderophore secretion compared to vehicle control treated bacterial cultures (Fig. 1B & C).

In separate studies we found that estrogen did not augment the planktonic growth of *P. aeruginosa* over an 8 h or 24 h period (Supplemental Fig. S2).

3.2. Estrogen increases motility of all P. aeruginosa isolates and biofilm formation of P. aeruginosa strain PsA01

To assess the effect of E2 on bacterial motility, *P. aeruginosa,* clinical CF isolates and strains were inoculated onto 0.25, 0.5 and 1.0% agar plates ± E2 or ethanol (vehicle control) for 24 h and swimming, swarming and twitching zone diameter were measured. E2 had no effect on the swimming motility of any of the strains tested (Fig. 2A). E2 did however significantly increase the swarming motility of all strains tested in 0.5% agar compared to vehicle controls. An increase in swarming of between 48 and 52 % was observed (P < 0.05) (Fig. 2B). A significant increase in the twitching motility of all isolates and strains tested was also observed with E2 treatment compared to vehicle controls, with increases of 61–111% in twitching in 1.0% agar, (P < 0.05, P < 0.01) (Fig. 2C).

The effects of E2 on *P. aeruginosa* biofilm formation were tested using a crystal violet assay performed in 96-well plates. E2 had no effect on biofilm formation of the early isolate, the late isolate or PA27853 compared to vehicle control wells, however, a significant increase in biofilm formation was observed for PAO1 (P < 0.001) (Fig. 2D). Testosterone (10 nM) had no effect on biofilm formation of PAO1

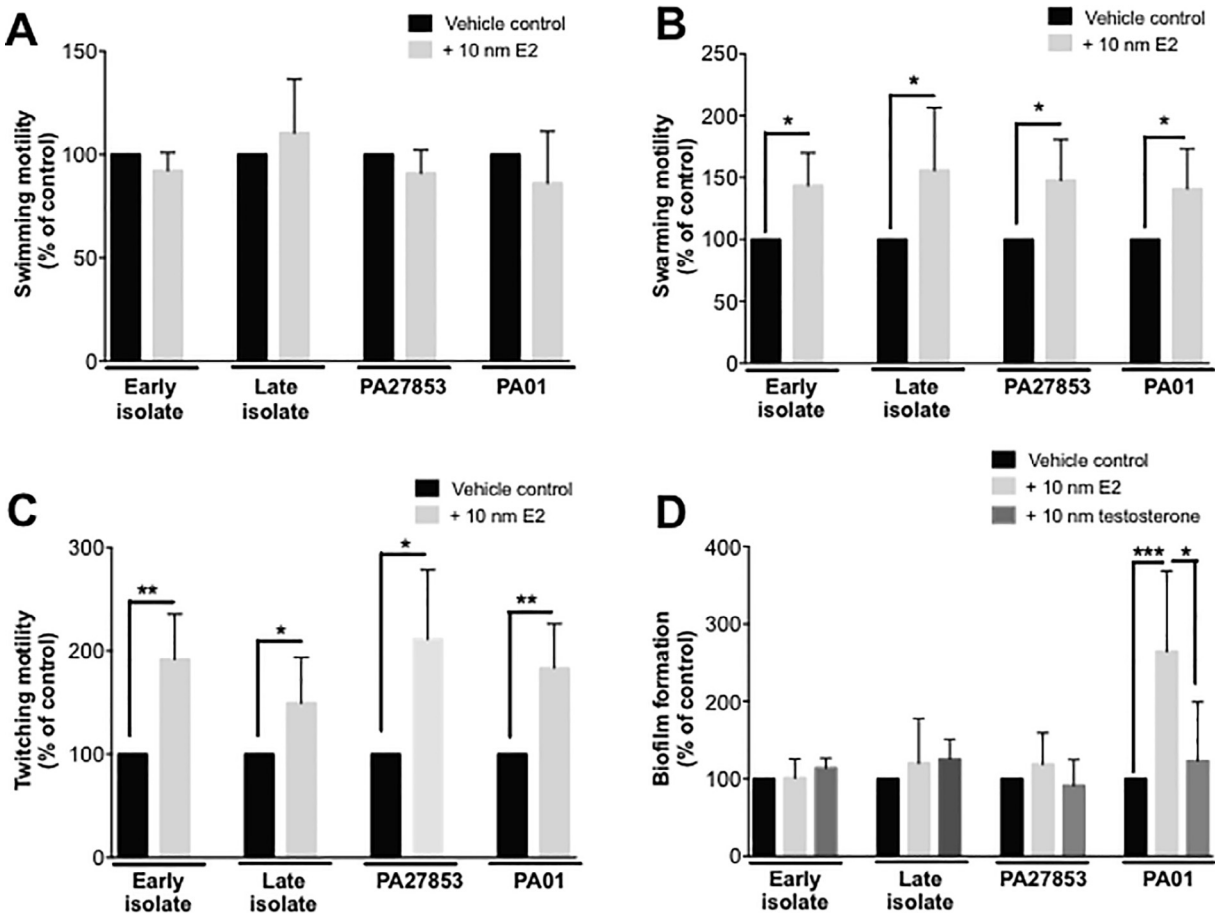

Fig. 2. Effects of 10 nM E2 on *P. aeruginosa* motility and biofilm formation. (A) swimming (0.25% agar), (B) swarming (0.5% agar), (C) twitching (1.0% agar), (D) biofilm formation. The results are presented as the mean ± SE of the percent distance travelled (motility) and absorption at 590 nm (biofilm) with respect to the corresponding control. Significance *P < 0.05, **P < 0.01 (vs vehicle control).

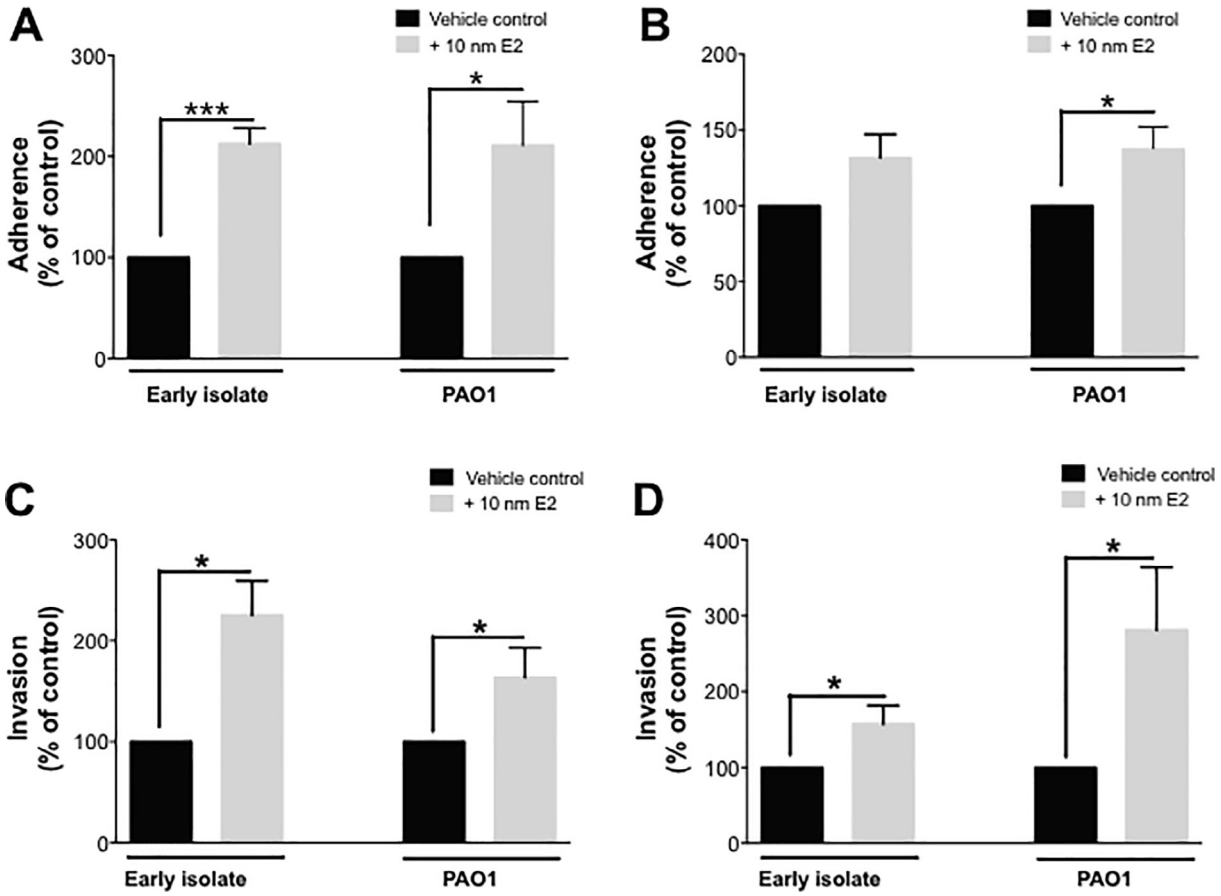

Fig. 3. Effects of 10 nM E2 on *P. aeruginosa* adherence and invasion into bronchial epithelial cells. (A) adherence to NuLi-1 cells, (B) adherence to CuFi-1 cells, (C) invasion in NuLi-1 cells, (D) invasion in CuFi-1 cells. The results are presented as the mean ± SE of the percent adherence/ invasion with respect to the corresponding control. Significance *P < 0.05, ***P < 0.001 (vs vehicle control).

(Fig. 2D).

3.2.1. Estrogen enhances P. aeruginosa adherence to and invasion into bronchial epithelial cells.

To investigate if E2 effected bacterial adherence and invasion into epithelial cells, NuLi-1 and CuFi-1 bronchial epithelial cells were infected with the early clinical CF isolate and PAO1 ± 10 nM E2 or ethanol (vehicle control) for 2 h as described by Higgins et al. [14]. Adherence of both the early clinical CF isolate and PAO1 was significantly increased in the presence of E2 compared to vehicle control in NuLi-1 cells ($P < 0.05$, $P < 0.001$) (Fig. 3A) with a greater increase in adherence observed in NuLi-1 cells (Fig. 3A). Adherence of the early clinical CF isolate was not significantly increased in CuFi-1 cells, however, adherence of PAO1 was significantly increased in the presence of E2 compared to vehicle controls ($P < 0.05$) (Fig. 3B). Invasion of both of these isolates into NuLi-1 and CuFi-1 cells was also significantly increased with E2 compared to vehicle controls ($P < 0.05$) (Fig. 3C & D).

3.3. The enhancement in P. aeruginosa invasion in the presence of estrogen is due to estrogen effects on P. aeruginosa and not on epithelial cells

To elucidate if the increase in *PsA* invasion in the presence of E2 was due to an effect of E2 directly on *P. aeruginosa* or on the epithelial cells, both the bacteria and the epithelial cells were treated separately with E2 or ethanol (vehicle control) for 72 h. E2 was then removed before the infection was carried out. *P. aeruginosa* treated with E2 and infected onto untreated epithelial cells displayed a significant increase in invasion levels compared to vehicle control treated cells ($P < 0.05$) (Fig. 4A & B). Epithelial cells treated with E2 and infected with untreated *P. aeruginosa* displayed no significant difference in invasion levels compared to vehicle controls (Fig. 4C & B).

3.4. Serms block estrogens effects on P. aeruginosa motility

Targeting the ability of *P. aeruginosa* to bind E2 is of utmost importance to inhibit an increase in virulence in female CF lung infections. To determine if the effect of E2 on enhancing *P. aeruginosa* virulence could be blocked by estrogen receptor (ER) antagonists, a twitching assay with the addition of selective estrogen receptor modulators (SERMs) was conducted on the early clinical CF *PsA* isolate. Two ER inhibitors, Tamoxifen and ICI 182780, and two ER activators, ERbeta agonist DPN and ERalpha agonist PPT, were tested ± E2 or appropriate vehicle controls, and plates were incubated for 24 h. As already observed in Fig. 2C, an increase in the twitching motility of the early clinical CF isolate with E2 was confirmed ($P < 0.01$) (Fig. 5A). However, when treated with the ER antagonists, tamoxifen or ICI 182780, the twitching motility of the early clinical CF *PsA* isolate was significantly inhibited in the absence or presence of E2 ($P < 0.01$, $P < 0.001$) (Fig. 5A). The ER*alpha* and ER*beta* selective agonists had no significant effect on the twitching motility of the early clinical CF isolate ± E2 (Fig. 5A). The effects of all SERMs tested are displayed in Fig. 5B.

3.5. Steroid selectivity of PsA virulence

To address the possibility of 'off-target' effects of estrogen we determined the steroid specificity of estrogen action by testing the effects

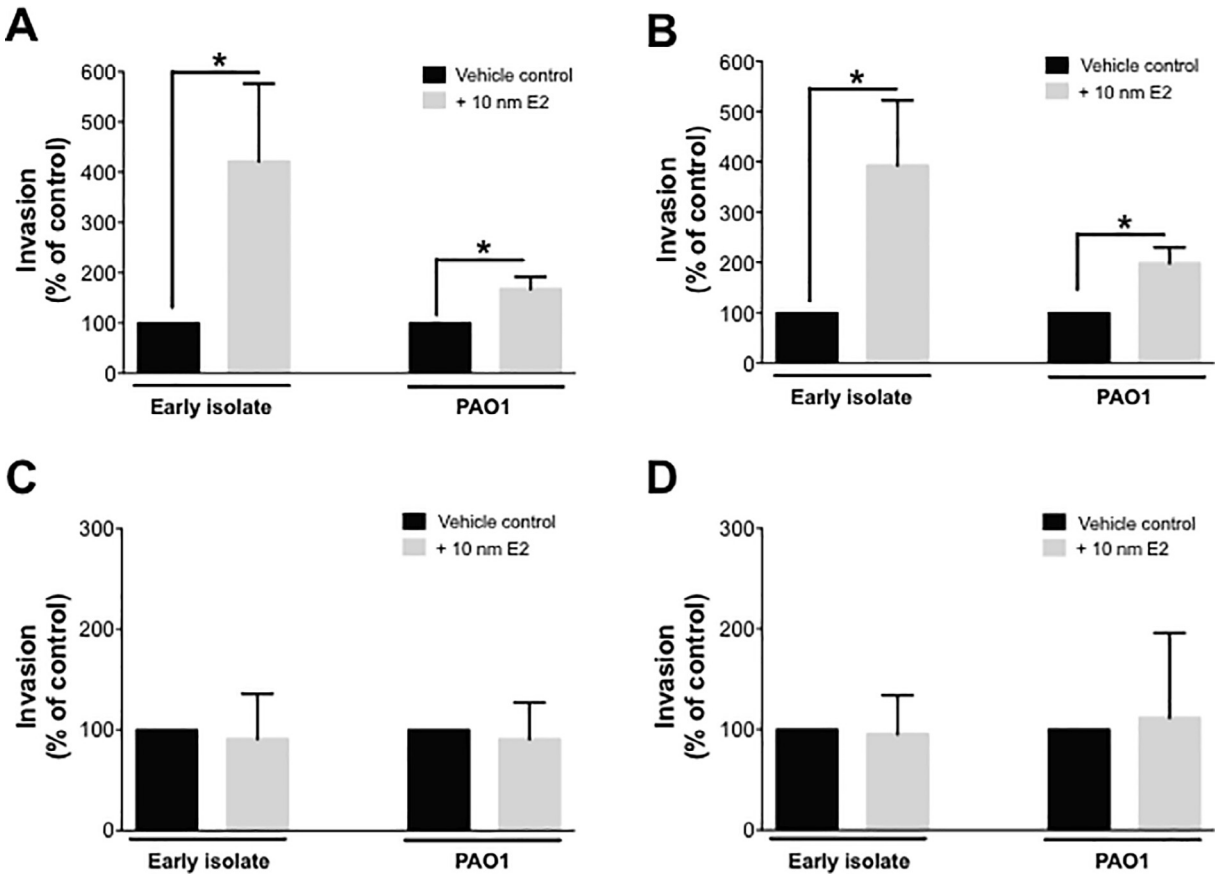

Fig. 4. Effects E2 (10 nM) treatment on *PsA* invasion when *P. aeruginosa* and epithelial cells were treated separately with E2. (A) NuLi-1 cells treated with E2, (B) CuFi-1 cells treated with E2, (C) *P. aeruginosa* treated with E2, NuLi-1 cells, (D) *P. aeruginosa* treated with E2, CuFi-1 cells. The results are presented as the mean ± SE of the percent invasion with respect to the corresponding control. Significance *P < 0.05, (vs vehicle control).

of a wide range of steroid hormones on *PsA* motility. The potency of various steroid hormones to stimulate motility of early clinical CF *PsA* isolate was in the order estradiol ≫ estrone > E3 estriol ≥ testosterone ≥ progesterone ≫ aldosterone, cortisol (Fig. 6). *PsA* expresses a highly selective estrogen binding protein (EBP) or estrogen receptor in its cytosol [24]. In Fig. 6, we compare the steroid potency to stimulate *PsA* motility with the steroid binding specificity of the *PsA* estrogen binding protein. The latter data were calculated from the competition of unlabelled steroid for [^{3}H]estradiol binding to the *PsA* cytsosolic EBP published in [24]. The data show excellent agreement between EBP steroid selectivity and potency to stimulate *PsA* motility, pointing to selective effects of estrogen on *PsA* virulence.

3.6. Estrogen inhibits ciliary beat dynamics

PsA virulence can be exacerbated by reduced mucociliary clearance. A key parameter determining the rate of mucus clearance is ciliary beat frequency (CBF). Estrogen E2 has been shown to rapidly reduce ASL height at physiological concentrations of 0.1–1 nM [7]. Since the power or intensity of ciliary beating is a more faithful measure of the

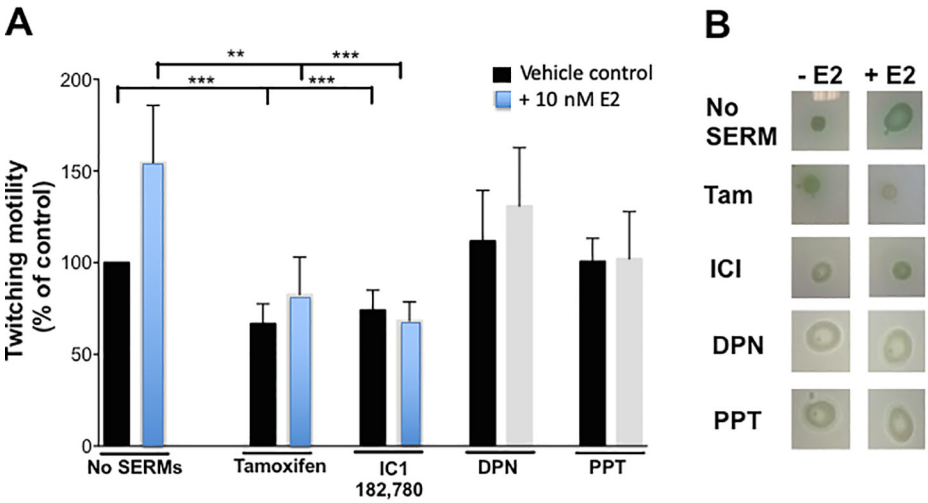

Fig. 5. Effects of SERMs on *P. aeruginosa* motility in the absence or presence of estrogen (10 nM). (A) *PsA* motility (represented as percentage of vehicle control) of the early clinical CF isolate treated with vehicle control or Tamoxifen (30 μM), ICI 182, 780 (0.5 nM), ERbeta agonist DPN (5 nM) or ERalpha agonist PPT (15 nM). (B) Representative images of twitching results for the indicated treatments. The results are presented as the mean ± SE of the percent distance travelled with respect to the corresponding control. Significance **P < 0.01, ***P < 0.001 (vs vehicle control ± E2).

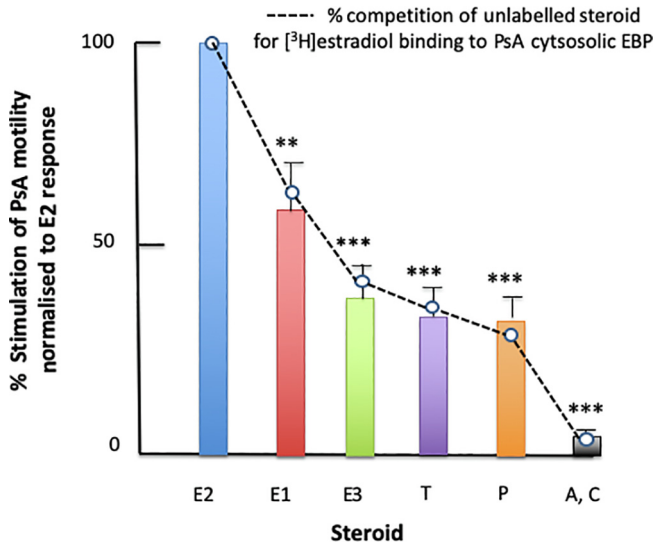

Fig. 6. Potency of various steroid hormones to stimulate motility of *P. aeruginosa* normalized to the maximum mean stimulatory response to estrogen. Steroids were added at a final concentration of 10 nM to the PsA growth medium 24 h before the motility assay. The results are presented as the mean ± SE of the percent stimulation in motility with respect to estrogen. Significance **P < 0.01, ***P < 0.001, experiments performed in triplicate (n = 6). Steroid legend key: E2 estradiol, E1 estrone, E3 estriol, T testosterone, P progesterone, A,C, aldosterone, cortisol. Dashed line is the % competition of unlabelled steroid for [^{3}H]estradiol binding to the PsA cytsosolic estrogen binding protein EPB calculated from data in (Rowland SS et al. J Steroid Biochem Mol Biol. 1992 Aug;42(7):721–7).

effectiveness of ciliary movement to drive mucociliary clearance, we determined the effects of estrogen on ciliary beat intensity (CBI). CBI was calculated from the product of the frequency in ciliary beat oscillations and the ciliary beat amplitude (CBA) measured from peak to peak changes in optical density (OD) of ciliary beating. We tested the effects of E2 on ciliary beat intensity in Nuli-1 and CuFi-1 epithelial monolayers in flooded cell culture wells to avoid indirect effects of E2 on CBF via ASL height changes. The effects of estrogen E2 (1 nM) on ciliary beating was tested in Nuli-1 and CuFi-1 cells after a control period (5 min) to establish a stable CBF. Estrogen rapidly decreased ciliary beating and intensity in both normal and CF cells (Figs. 7 and 8). Under control conditions, the ciliary beat frequency in NuLi-1 bronchial epithelial cells varied over a range of 15–20 Hz with a mean CBF = 18 ± 4 Hz, CBA = 140 ± 30 OD a.u. , CBI = 2,520 ± 120 Hz.OD a.u., n = 6. Estrogen reduced ciliary beat amplitude and intensity without appreciably affecting ciliary beat frequency in normal epithelium. Following 15 min treatment of NuLi-1 cells with E2, the mean CBF = 15 ± 4 Hz, CBA = 107 ± 8 OD a.u., CBI = 1610 ± 212, n = 6. The ciliary beat frequency in CF epithelium was in a similar range as normal epithelium, however, the amplitude and intensity of ciliary beating was very much attenuated, even under flooded ASL conditions, indicating that CF epithelia have a low intrinsic CBF independent of ASL height. In CuFi-1 cells, the CBF varied over a range of 12–20 Hz, mean CBF = 15 ± 5 Hz, CBA = 75 ± 10 OD a.u., CBI = 1,125 ± 85, n = 6. In CF epithelia, estrogen produced a rapid decrease in ciliary beat oscillations within 2 min, reaching a steady state of practically no observable ciliary beating after 20 min (Fig. 7). Following 15 min treatment of CuFi-cells with E2, the mean CBF = 12 ± 4 Hz, CBA = 27 ± 8 OD a.u., CBI = 325 ± 95, n = 6. These data are summarised in Fig. 8. Taken together, these experiments show that E2 inhibits ciliary beat dynamics in normal and CF airway epithelia and is consistent with what was observed previously for E2 reduction in ASL height [7]. It is expected that the reduction of ciliary beat dynamics and ASL height by estrogen will contribute to a reduced mucociliary clearance and increased trapping of virulent *PsA* to exacerbate infection and inflammation in the CF airway.

4. Discussion

Estrogen (E2) exacerbation of CF lung disease has been well-described [3,6,7,32,33] and the E2 effects on *Pseudomonas aeruginosa* virulence (mucoidy phenotype) in CF was first published by our group in 2012 [6]. In this paper, we go further and describe the effects of E2 to enhance *PsA* motility and invasion through direct action of the sex hormone on the microbe independently of the epithelial host. The *PsA* virulence effect of E2 is not unique to the CF host as it can occur in a petri dish, however, the E2 effects on *PsA* virulence are additive to the exacerbating effects of E2 on lung physiology in CF (reduced airway surface liquid height, ENaC activation and hyperabsorption of Na^+, inhibition of non-CFTR mediated Cl^- secretion), (Supplemental Fig. S3), We also found additional exacerbating effects of estrogen on mucociliary clearance capacity by inhibition of ciliary beat. Thus E2 exacerbation of *PsA* virulence becomes particularly relevant in CF lung disease. Microbial endocrinology is an emerging field and several reports have been published indicating an effect of sex hormones on *P. aeruginosa* phenotype. This paper reports for the first time, the direct effects of E2 on *P. aeruginosa* virulence factors and interactions with the host epithelium.

In our study, we found that E2 did not promote *P. aeruginosa* proliferation *in vitro* and this was observed for the two clinical CF isolates, PAO1 and PA27853 tested, indicating that *P. aeruginosa* does not utilise estradiol for proliferation (*PsA* can use steroids as a carbon source for growth). E2 can increase bacterial numbers in some species, although the underlying mechanisms are poorly understood. *Prevotella nigrescens* and *P. pallens* numbers in planktonic form were increased with estradiol [16], and estradiol and progesterone reportedly stimulated the growth of *lactobacillus* spp, *Escherichia coli* and *Streptoccocus* spp. [17].

P. aeruginosa secreted virulence factors play an important role in enhancing microbial survival within the CF lung. Pyocyanin is a redox activated phenozine toxin secreted by *P. aeruginosa* and it is found in high concentrations in the sputum of CF patients [18]. This toxin is involved in acute infection and has the ability to damage the cilia of respiratory epithelium [18], increase IL-8 secretion by epithelial cells, induce apoptosis and inhibit T-cell proliferation [19]. We found that E2 significantly increased the levels of pyocyanin secretion in both of the clinical CF isolates used in our study compared to vehicle controls. The same effect was not observed in the lab strains PAO1 and PA27853. The difference in the effect of E2 on pyocyanin production between the clinical isolates and laboratory strains is possibly due to the different culturing environments. The increase in pyocyanin production in the clinical CF isolates may more closely represent the microbiology in the CF lung as PAO1 and PA27853 have had time to adapt to *in vitro* growth and can act differently over time compared to clinical isolates.

P. aeruginosa proteases such as elastase B, alkaline protease, protease IV and PasP can be detected in the lungs of CF patients [20] and breach epithelial barriers to aid cell invasion and gain access to the blood stream [21]. We observed no alteration in secreted protease levels by any of the isolates exposed to E2, suggesting that estradiol does not affect protease secretion by *P. aeruginosa in vitro*. This is the first report of the effect of estradiol on bacterial protease secretion and it is possible that *P. aeruginosa* does not need this virulence factor to be altered to gain a survival advantage during infection.

Siderophores are secreted iron chelating molecules and the ability of *P. aeruginosa* to acquire iron within the host is essential for bacterial growth and infection [22]. *P. aeruginosa* secretes two siderophores, pyoverdine and pyochelin and the absence of these siderophores severely retards growth and decreases virulence. Both the clinical CF isolates and the lab strains PAO1 and PA27853 produced siderophores under iron deprivation, however E2 did not alter siderophore levels of *P. aeruginosa* as indicated by the CAS assay. There are no previous

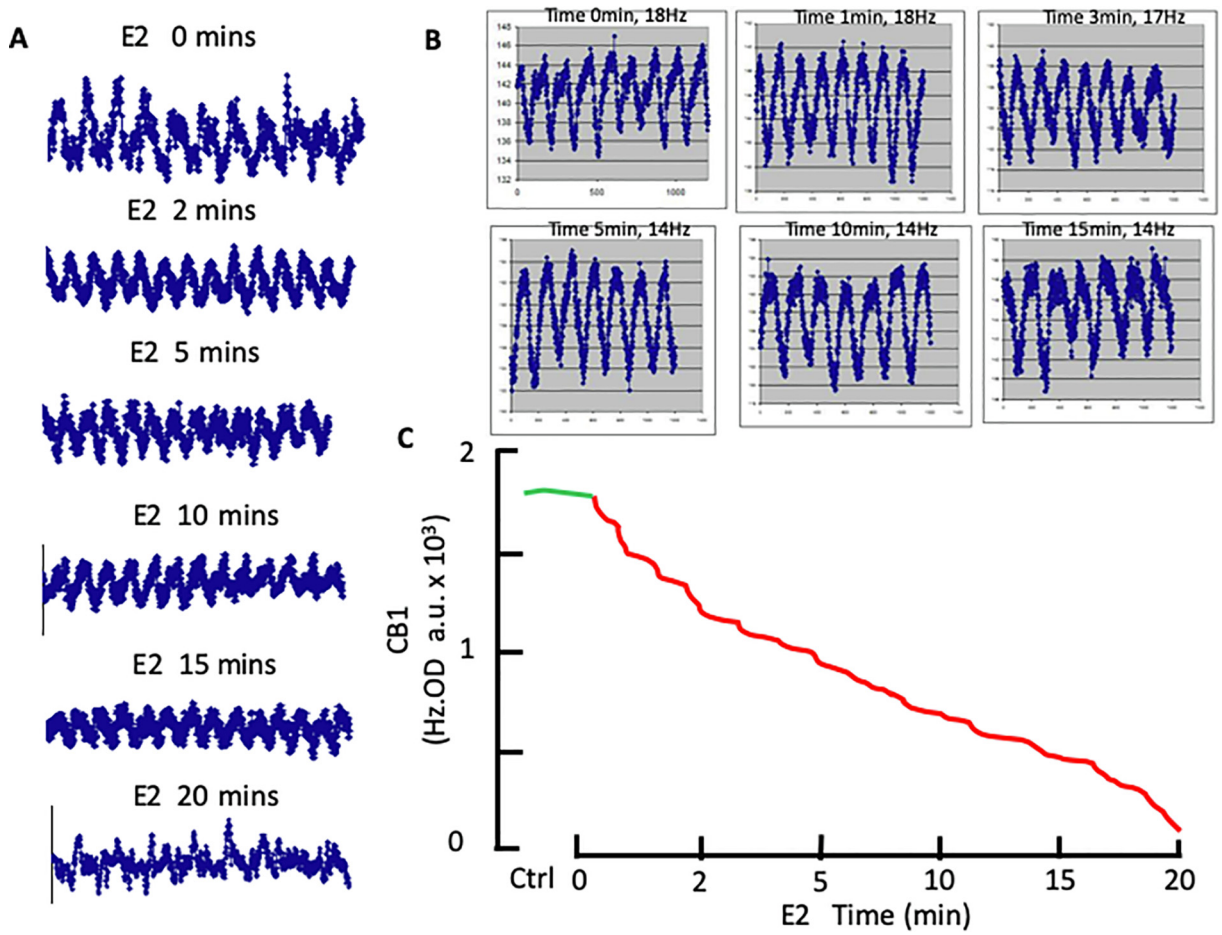

Fig. 7. Estrogen effects on ciliary beating in CuFi-1 CF airway epithelial monolayers grown in flooded Millicell culture wells and imaged with a Zeiss CLSM 510 confocal microscope and analysed using ImageJ software®. The spontaneous ciliary beat frequency (CBF) showed dyskinesia in CF cells (A) when compared with normal airway cells showing rapid CBF of 14–18 Hz (B). Estrogen produced a rapid inhibition of ciliary beat intensity (A, C) which collapsed to undetectable levels within 20 min of hormone treatment.

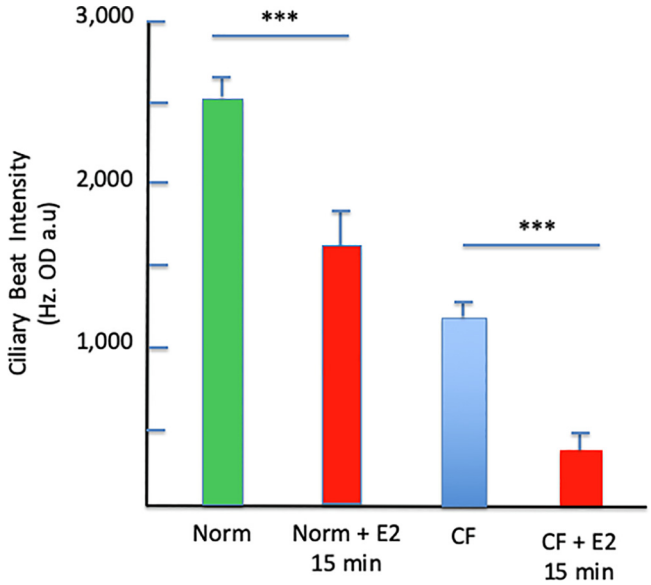

Fig. 8. Analysis of ciliary beat intensity (CBI) in normal NuLi-1 and CF CuFi-1 bronchial epithelial cells under control conditions and after 15 min exposure to estrogen (1 nM). CBI was calculated from the product of the frequency and peak to peak amplitude of ciliary oscillations. Significance *** $P < 0.001$, $n = 6$.

reports on the effects of E2 on bacterial siderophores. *P. aeruginosa* has numerous transport mechanisms, such as Haem uptake pathways [23], to acquire iron and an increase in siderophore production may be a redundant and unnecessary use of energy.

Bacterial motility is essential in the colonization of surfaces by *Pseudomonas* and the subsequent formation of biofilms. *P. aeruginosa* uses swimming motility, which is mediated by the flagella and the surface-associated swarming and twitching motilities, which are predominantly mediated by hyper-flagellation and type-IV pili, respectively to colonize new surfaces. We observed no alteration in *P. aeruginosa* swimming in the presence of E2, however, swarming and twitching motility of clinical CF isolates and laboratory strains were significantly increased with E2. This suggests that E2 is possibly affecting *Pseudomonas* type IV pili and hyper-flagellation resulting in increased motility. An increase in biofilm formation of PAO1 was observed with E2 but this effect was not observed for the other isolates. It is possible that PAO1 has adapted over the years during culturing in the laboratory environment resulting in a different biofilm response to estrogen. The *Pseudomonas* PAO1 biofilm responses appear to be specific to the female sex hormone as testosterone was without effect in PAO1. We have previously reported that an increase in alginate production with estradiol by *P. aeruginosa* was not observed with testosterone [6] suggesting that *P. aeruginosa* does not bind androgens to enhance virulence.

A constitutive estrogen-binding protein (EBP receptor) with high selectivity for estrogen over other steroids has been described in the cytosol of *Pseudomonas aeruginosa* [24]. The steroid binding specificity

of the EBP, in order of decreasing affinity, was estradiol, estrone, dihydrotestosterone, estriol, testosterone, progesterone and promegestone. Other steroid hormones such as aldosterone and cortisol tested did not compete for estradiol binding. The steroid selectivity of the EBP agrees well with the steroid potency we observed to stimulate *PsA* motility in the order estradiol ≫ estrone > E3 estriol ≥ testosterone ≥ progesterone ≫ aldosterone and cortisol. The excellent agreement between EBP steroid selectivity and potency to stimulate *PsA* motility points to selective effects of estrogen on *PsA* virulence. We have previously demonstrated that estrogen causes mucoid conversion of *PsA* which contributes to sexual dimorphism in CF lung disease severity [6]. It was concluded that estrogen binding to the EBP receptor in *PsA* cytosol is the most likely transduction mechanism for the E2 induced frameshift mutation in mucA, a key regulator of alginate biosynthesis in *P. aeruginosa.*

P. aeruginosa can bind to and enter into epithelial cells to gain access to the blood stream [25]. Adherence to epithelial cells is mediated by a number of adhesions, including type IV pili and lipopolysaccharide [26]. We observed a significant increase in *P. aeruginosa* adherence to normal and CF lung cells with estrogen treatment. One other study has attempted to discriminate between pathogen and host interactions with E2 in which it was observed that E2 enhanced adherence of *C. trachomatis* to HeLa cells post-infection and not in pre-infection models [27]. This report suggested that estradiol was acting on the HeLa cells and not *C. trachomatis,* although no ER was detected. In our study, we further elucidated host-pathogen effects of E2. A *P. aeruginosa* clinical isolate and laboratory strain were grown alone with E2 or vehicle control and epithelial cells were also treated separately with E2 or vehicle control and an invasion assay was carried out. When the epithelial cells alone were treated with E2 there was no significant increase in invasion by *P. aeruginosa*. However, when *P. aeruginosa* alone were treated with E2 and added to the untreated cells, a significant increase in invasion into both normal and CF bronchial epithelial cells was observed. These findings suggest that E2 acts directly on the bacteria to enhance virulence and possibly effecting adhesion factors such as Type IV pili. Adhesion and invasion by *P. aeruginosa* into the airway epithelium could be facilitated by a reduction in ciliary beat, lowered ASL height and a compromised mucociliary clearance [28].

In general, males tend to fare worse than women in response to infection. We have reviewed the epidemiology of sex differences in infection rates [29] and found that this sex difference applies essentially to methicillin-resistant Staphylococcus aureus (MRSA) carriage and bloodstream infection (BSI), which shows a male predominance. Estrogen effects on *PsA* virulence is unique to CF lung disease for a multitude of reasons. Firstly, CF patients have much higher *PsA* colonization rates than the general population and this is increased in females over males at puberty [3,30]. There are several reasons for this CF virulence, primarily CFTR mutations allow higher airway adherence and invasion to *PsA*. Moreover, microbial virulence is exacerbated by high estrogen states in females due to estrogen transformation of *PsA* into a mucoid phenotype and alginate production causing *PsA* to stick to CF airways in a biofilm and resist antimicrobial attack [6]. Estrogen also inhibits the innate mucosal defence against *PsA* invasion and adherence in CF airways [31]. Furthermore, estrogen compromises ASL dynamics leading to a favorable environment for bacterial proliferation and sustained inflammation resulting in epithelial lung tissue injury, fibrosis and remodelling [32,33].

Bloodstream infection (BSI) was first identified in association with cystic fibrosis 40 years ago [34,35]. Fortunately, it is now quite rare to see CF patients with bacterimia or septicaemia except in severe febrile pulmonary exacerbations (FPE) with *Burkholderia cepacia* [15]. Moreover, the majority (82%) of significant blood stream infections originate from totally-implantable vascular access devices (TIVADs); with only 9% attributed to the lower respiratory tract [36]. The very few clinical studies of bloodstream infections in CF patients did not address sexual dimorphism and point to very low rates of bacterimia. The very first study to examine the incidence of positive findings from blood cultures collected during FPE among CF patients reported an incidence of 3.5%. [37]. A recent prognostic study showed a BSI in CF patients of 0.8% with suspected sepsis in febrile pulmonary infection [38] and a more recent large-scale retrospective study has revealed female sex as a risk factor in mortality of patients with *Pseudomonas aeruginosa* bacteraemia (42).

The low rates of BSI in CF patients does not mean that microbial invasion into airway epithelial cells is not an important factor in CF lung exacerbations, for many microbial species use epithelial invasion to evade the innate immune response to cause virulent inflammation of the CF airways [39]. We have previously shown that pro-resolution anti-inflammatory mediators can reduce *PsA* invasion in CF airway epithelia [40]. Moreover, the estrogen effects on *PsA* cellular invasion observed in our study are most likely a compounding factor in the sexual dimorphism of lung function exacerbations observed in high circulating plasma estrogen. In fact, CF lung exacerbations are not only sex-dependent but also show variations in severity during the menstrual cycle [6]. Increased proportions of mucoid bacteria were isolated from CF female lung lavage during exacerbations occurring in the follicular phase (high plasma estradiol), with a variable *P. aeruginosa* phenotype evident *in vivo* during the course of the menstrual cycle corresponding to fluctuating estradiol levels.

Limiting estrogen availability to *P. aeruginosa* could potentially compromise pathogen motility, biofilm formation and interactions within the host. Given that the *PsA* estrogen binding protein has high affinity for E2, targeting this receptor protein could lead to the inability of *P. aeruginosa* to utilize estradiol to enhance virulence. We used SERMs to investigate their ability to block estrogen effects on *P. aeruginosa* virulence factors and observed a significant decrease in twitching motility in response to estrogen treatment when grown with estrogen receptor antagonists tamoxifen or IC1 182780, suggesting that these SERMs are binding to the *PsA* EBP to block the estrogen responses. Even without E2 treatment, the twitching motility of *P. aeruginosa* was significantly hampered by tamoxifen or ICI 182780 and it may be possible that other virulence factors in the culture media are bound by the EBP and inhibited by SERMs. The absence of effects of the ER*alpha* and ER*beta* agonists PPT and DPN on twitching motility may indicate that the *PsA* estrogen binding protein is structurally distinct from mammalian ER. Gel filtration of the native EBP protein gave an estimated molecular weight of 215 kDa and a Stokes radius of 50.2 Å [24], compared to a mW for the estrogen receptors ERα 66 kDa and ERβ 55 kDa, and Stokes radius of 44 Å.

Another mechanism by which estrogen may exacerbate *PsA* virulence is to reduce mucociliary clearance of *PsA* pathogens and this is predicted from the inhibitory effect of estrogen on airway surface liquid height [7,33]. The inhibitory effects of estrogen on ciliary beat intensity independent of ASLh variations shown in this study indicate direct effects of estrogen on cellular and molecular mechanisms regulating ciliary beat dynamics.

5. Conclusions

Although there have been remarkable advances in CF therapeutics in the last decade, there still exists significant sex differences in morbidity and mortality in adult CF patients [41]. Microbial endocrinology is fast becoming a hot topic in CF research. Recent clinical studies have shown female gender as a significant risk factor for mortality in *Pseudomonas aeruginosa* bacteremia [42]. More studies to evaluate the molecular mechanisms behind sex hormones in pathogen virulence are needed to further understand the sexual dimorphism of CF infections. In this study we have demonstrated that estrogen augments *P. aeruginosa* virulence factors and its interactions with the host epithelium which can be inhibited by SERMs. Our work suggests that SERMs could be used as a potential treatment to reduce estrogen enhanced *P. aeruginosa* infections in females with CF.

CRediT authorship contribution statement

Jean Tyrrell: Data curation, Formal analysis, Writing - original draft. **Brian J. Harvey:** Conceptualization, Data curation, Formal analysis, Validation, Funding acquisition, Investigation, Project administration, Supervision, Writing - original draft, Writing - review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.steroids.2019.108575.

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